

$$\begin{aligned}\Leftrightarrow \sqrt{9-x^2} &= 2x-3 \\ \Leftrightarrow 9-x^2 &= (2x-3)^2 = 4x^2+9-12x \\ \Leftrightarrow 5x^2-12x &= 0 \Leftrightarrow 5x^2=12x \\ \Leftrightarrow x &= \frac{12}{5}.\end{aligned}$$

147. Let the total number of camels be x .

$$\begin{aligned}\text{Then, } x - \left(\frac{x}{4} + 2\sqrt{x}\right) &= 15 \Leftrightarrow \frac{3x}{4} - 2\sqrt{x} = 15 \\ \Leftrightarrow 3x - 8\sqrt{x} &= 60 \Leftrightarrow 8\sqrt{x} = 3x - 60 \\ \Leftrightarrow 64x &= (3x-60)^2 \Leftrightarrow 64x = 9x^2 + 3600 - 360x \\ \Leftrightarrow 9x^2 - 424x + 3600 &= 0 \Leftrightarrow 9x^2 - 324x - 100x + 3600 = 0 \\ \Leftrightarrow 9x(x-36) - 100(x-36) &= 0 \Leftrightarrow (x-36)(9x-100) = 0 \\ \Leftrightarrow x &= 36. \quad \left[\because x \neq \frac{100}{9} \right]\end{aligned}$$

148.

1	1	79	56	(134
	1			
23		79		
		69		
264		10	56	
		10	56	
				x

$\therefore$ Number of rows = 134.

149.

1	1	10	25	(105
	1			
20		10		
		0		
205		10	25	
		10	25	
				x

$$\text{Number of rows} = \sqrt{10914+111} = \sqrt{11025} = 105.$$

150. Let the number of girls in the group be x .
Then, number of oranges given to each girl = $2x$.
 $\therefore x \times 2x = 1250 \Leftrightarrow 2x^2 = 1250 \Leftrightarrow x^2 = 625 \Leftrightarrow x = \sqrt{625} = 25$.

151.

1	3	65	81	(191
	1			
29		2	65	
		2	61	
381		4	81	
		3	81	
		1	00	

$\therefore$ Number of men left = 100.

152. Money collected = (59.29×100) paise = 5929 paise.

$\therefore$ Number of members = $\sqrt{5929} = 77$.

153. Clearly, the required number must be a perfect square. Since a number having 8 as the unit's digit cannot be a perfect square, so 1,21,108 is not a perfect square.

154. $148877 = 53 \times 53 \times 53$

$$\therefore \sqrt[3]{148877} = 53.$$

53	148877
53	2809
	53

155. $681472 = 8 \times 8 \times 8 \times 11 \times 11 \times 11 = 8^3 \times (11)^3$.

8	681472
8	85184
8	10648
11	1331
11	121
	11

$$\sqrt[3]{681472} = 8 \times 11 = 88.$$

156. $262144 = 8 \times 8 \times 8 \times 8 \times 8 \times 8 = 8^6$.

$$\therefore \sqrt[3]{262144} = 8^2 = 64.$$

$$\text{Let } 1728 + \sqrt[3]{262144} \times x - 288 = 4491.$$

$$\text{Then, } 1728 + 64 \times x - 288 = 4491$$

$$\Leftrightarrow 27x = 4779$$

$$\Leftrightarrow x = \frac{4779}{27} = 177.$$

8	262144
8	32768
8	4096
8	512
8	64
	8

157. Let $99 \times 21 - \sqrt[3]{x} = 1968$.

$$\text{Then, } 2079 - \sqrt[3]{x} = 1968 \Leftrightarrow \sqrt[3]{x} = 2079 - 1968 = 111$$

$$\Leftrightarrow x = (111)^3 = 1367631.$$

$$\begin{aligned}158. (.000216)^{1/3} &= \left(\frac{216}{10^6}\right)^{1/3} \\ &= \left(\frac{6 \times 6 \times 6}{10^2 \times 10^2 \times 10^2}\right)^{1/3} = \frac{6}{10^2} = \frac{6}{100} = .06.\end{aligned}$$

$$159. \sqrt[3]{4\frac{12}{125}} = \sqrt[3]{\frac{512}{125}} = \left(\frac{8 \times 8 \times 8}{5 \times 5 \times 5}\right)^{1/3} = \frac{8}{5} = 1\frac{3}{5}.$$

$$160. \sqrt{.000064} = \sqrt{\frac{64}{10^6}} = \frac{8}{10^3} = \frac{8}{1000} = .008.$$

$$\therefore \sqrt[3]{\sqrt{.000064}} = \sqrt[3]{.008} = \sqrt[3]{\frac{8}{1000}} = \frac{2}{10} = 0.2.$$

161. $864 = 3 \times 3 \times 3 \times 4 \times 4 \times 2$.

Clearly, 864 when multiplied by 2 will become a perfect cube.

Hence, $n = 2$.

4	864
4	216
3	54
3	18
3	18
3	6
	2